



SOIL BIOLOGICAL PROCESSES

DECOMPOSITION OF ORGANIC MATTER



The major fungi involved in weathering – taxonomic and ecological perspectives

In the fungal phyla Ascomycota, Basidiomycota, and Zygomycota, important functional groups of fungi with potential weathering capabilities can be recognized.



Lichen-forming fungi. Lichens, associations between fungi and algae or cyanobacteria, date at least from the early Devonian (400 million years ago).

Lichens have played an important role in rock weathering and soil development (Chen *et al.* 2000).

Because the associated fungus is closer to the mineral interface than the plant, the weathering ability of lichens is most probably due mainly to the fungus.

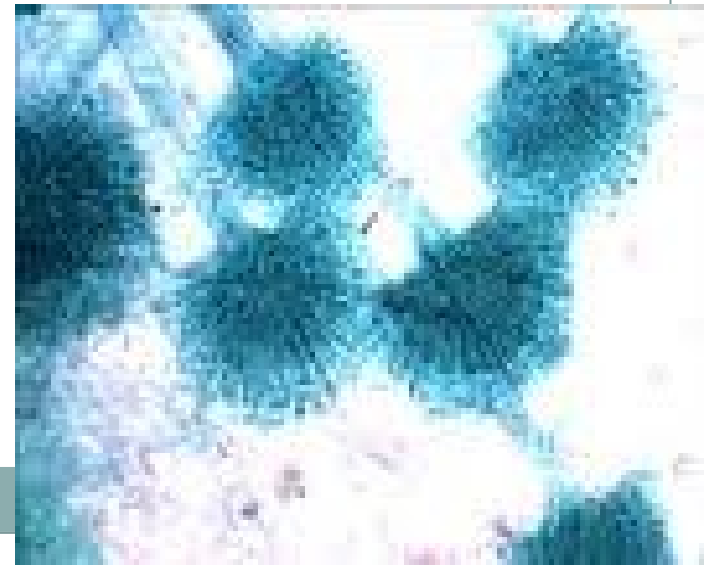
Mycorrhizal fungi, especially ectomycorrhizal (sheathing mycorrhizal) and ericoid mycorrhizal fungi, which were recognized recently as potentially important for mineral weathering (Lndeweert *et al.* 2001).

Ectomycorrhizae on roots of southern yellow pine

Roots of saplings (aged 10-15) in a long-term study of forest succession



Saprotrophic fungi, *Aspergillus niger* which use relatively simple carbohydrate sources and quickly proliferate under favorable conditions.





Saprotrophic organisms that use the lignocellulose complex.

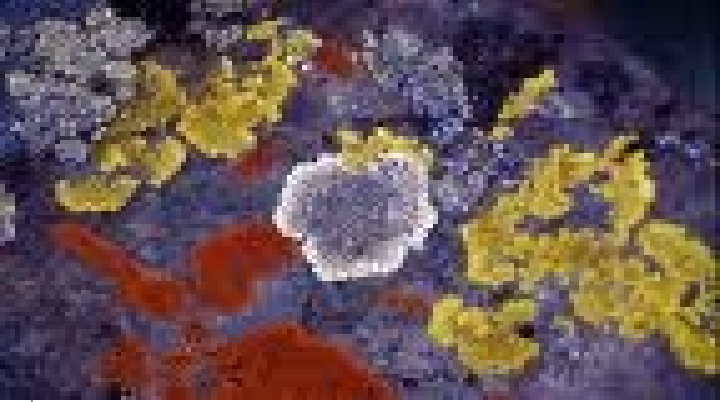
White rot and brown rot fungi produce large amounts of organic anions (especially oxalate) that they use for generating hydrogen peroxide and for the production of Mn chelates that can oxidize a range of phenolic compounds may therefore be potential weathering agents.



How do fungi weather rocks and minerals?

Fungi can penetrate solid materials using both physical and chemical tools. The mycelial growth form allows fungi to exploit weak spots on the rock or grain surface.





Lichen compounds. “Lichen acids”, produced by the fungal

partner in the lichen symbiosis, have for a long time been recognized as the major weathering components of lichens growing on rocks.

Furthermore, lichen compounds are usually concentrated in the upper thallus, at the fungal-algal or cyanobacterial interface, not at the mineral interface.

In the field, metal-lichen-acid complexes have very rarely been reported, suggesting they play a limited role in weathering.



Decomposition and soil organic matter





Plant Litter Decomposition

phases:

Leaching by water. Removes soluble materials (nutrients, simple C

Fragmentation by soil animals.

Chemical alteration. Changes chemical composition of detritus.

organic matter molecules

Compounds are decomposed at different rates,
new compounds are synthesized

Microbial activity controlled by:

Environment: soil T, H₂O, soil O₂

Substrate: chemical makeup of organic matter

Rate of microbial activity determines rate of C
loss, nutrient release etc.

Rate of activity measured as organic matter mass
loss.

Fungi

Accounts for most aerobic decomposition

- 60-90% of microbial biomass in forests
- About half of microbial biomass in grassland

Broad enzymatic capability

- Cell walls (lignin, cellulose, hemicellulose)
- Cell contents (proteins, sugars, lipids)

Can transport metabolites through hyphae

- Surface litter (import nitrogen from soil)
- Wood degraders (import nitrogen from soil)

- **Bacteria**

Grow rapidly on labile substrates

Some function anaerobically

Dependent on substrates that diffuse. Diffusion gradient caused by

- Production of soluble substrates (enzymes)

- Uptake of substrates by bacterium

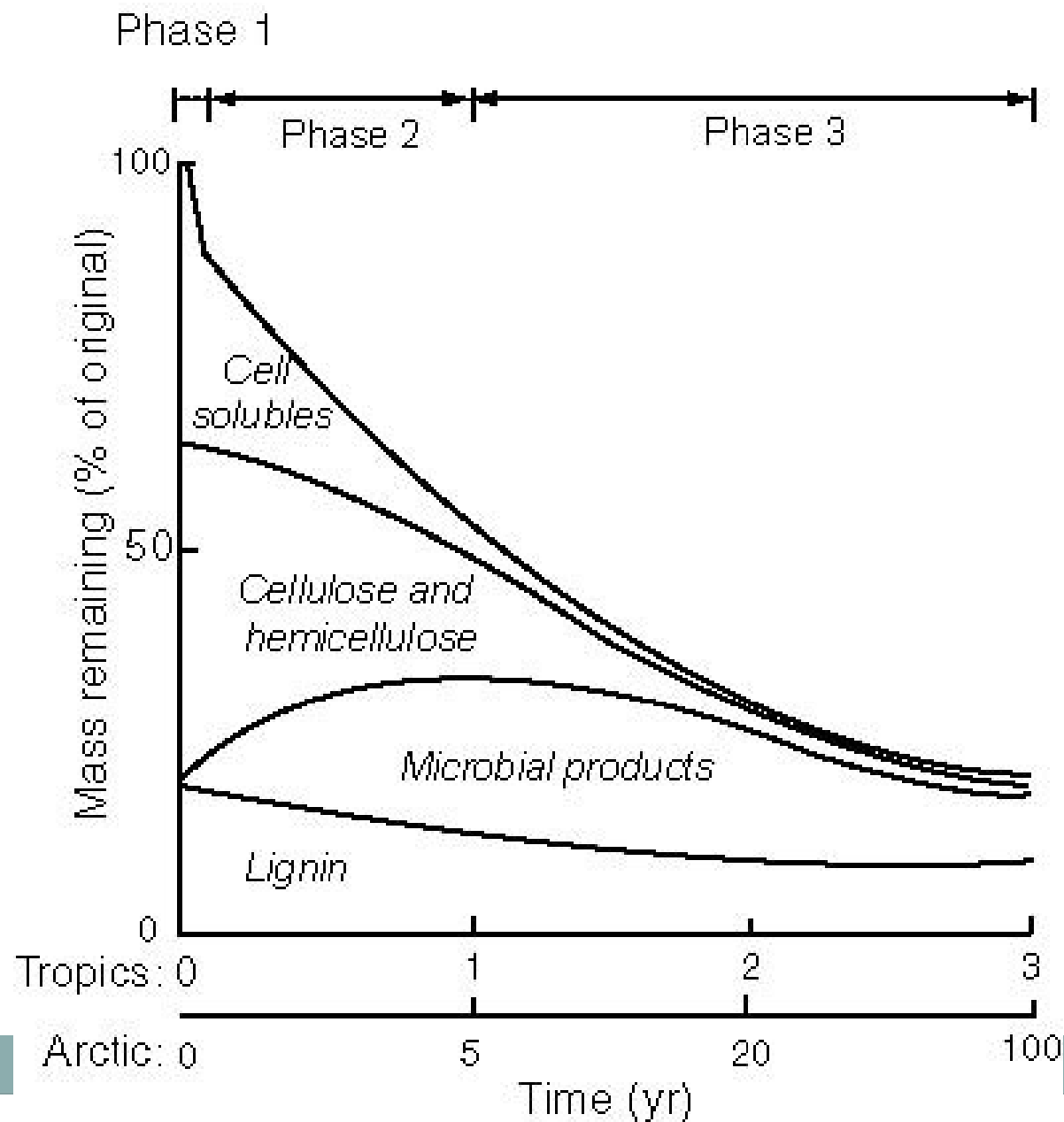
Spatial specialists

- Rhizosphere, macropores, interior of aggregates

- form biofilms on particle surfaces

Chemical specialists

- Different bacteria produce different enzymes



Organic matter in the soil

Organic matter is anything that contains carbon compounds that were formed by living organisms. Three main components are

Dead forms of organic material - mostly dead plant parts (85%)

living parts of plants - mostly roots (10%)

living microbes and soil animals

Partly decayed organic matter is called humus

continuum of organic matter decomposition from fresh
litter to well decomposed humus

Chemical Separation

Fulvic acids- soluble in acid, base

Humic acids- soluble in base, not acid

Humins - not soluble in base or acid

Major steps in the degradation of organic matter

Involves 4 major steps

Breakdown of compounds that are easy to decompose (e.g. sugars, starches and proteins)

Breakdown of compounds that take several years to decompose (cellulose and lignin)

Breakdown of compounds that take 10 years (e.g. waxes and phenols)

Compounds that takes 100-1000's of years (e.g. humus like substances, which are very complex)

Microbial degradation of organic matter

Involves complex processes. Chemical alteration of organic matter, physical fragmentation and release of mineral nutrients.

The dead organic matter is colonized by microbes and degraded with help of microbial enzymes.

Macromolecules are broken down into simpler units and further degraded into constituent elements.

Microbial degradation of organic matter..

Why do stems and leaves degrade at different rates?

Because they are made up of different kinds of molecules and require different enzymes.

E.g. Cellulose, predominant in leafy tissues are easily degradable than lignin, major component of wood.

Mineralization of organic matter

Mineralisation is the biological process where organic compounds in organic matter are chemically converted by the microorganisms in soil to simpler compounds, other organic compounds or mineralised nutrients.

Examples of tropical and temperate region plant-soil systems

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Tropical soil

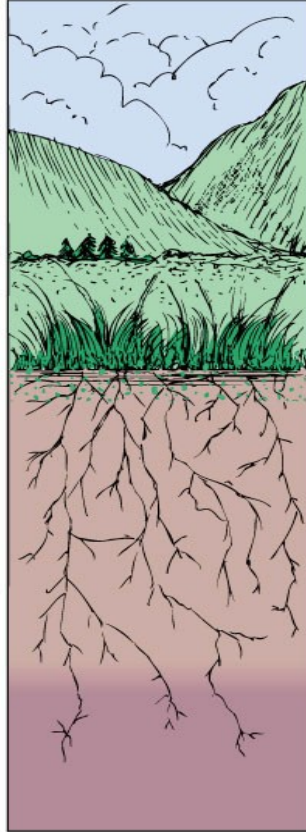


Rapid litter decomposition

Low organic matter soil

Roots penetrate into the litter layer to absorb nutrients, which minimizes losses by leaching

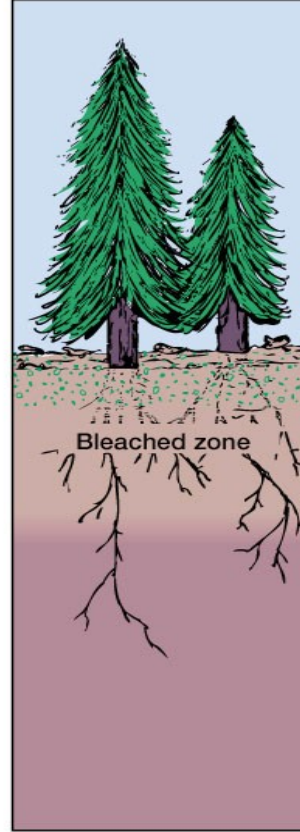
Temperate grassland



Litter can accumulate on surface

Deep root penetration and organic matter accumulation

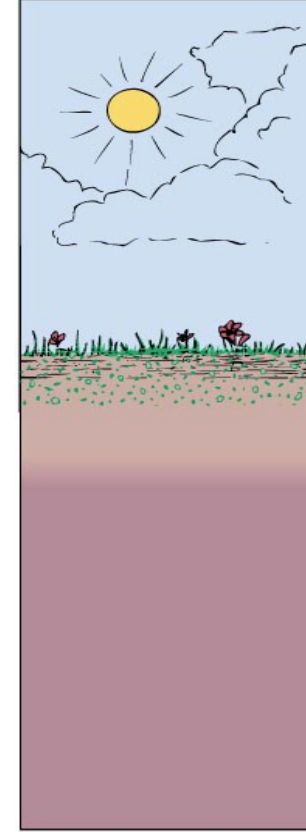
Temperate coniferous



Moist litter layer with lower oxygen levels

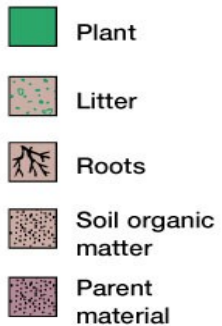
Acidic products accumulate and leach into the underlying soil, resulting in the formation of a bleached zone

Bog soil



Litter accumulation due to lower soil oxygen levels and decreased degradation

Formation of peat



Humification

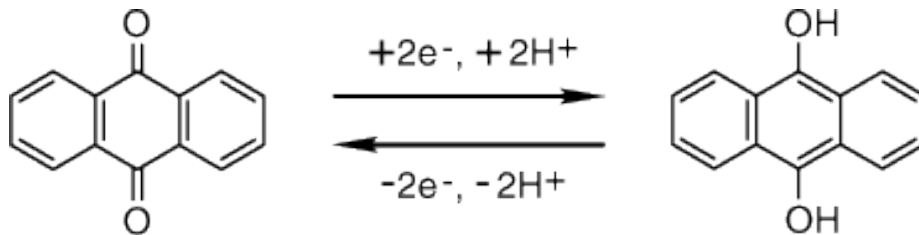
Selective preservation -- plant waxes, cutins, suberin, lignin, chitin, microbial cell walls are hard to decompose & become enriched in soil

Microbial transformations -- enzymatic breakdown by microbes produces low molecular weight soluble compounds such as amino acids.

Polyphenol formation --phenols in soil are derived from lignin breakdown, soil microbial synthesis, and by plants as defensive compounds

Humification

4. **Quinone formation** -- conversion of polyphenols to reactive quinones as a result of extracellular enzymes



5. **Abiotic condensation** -- quinones react spontaneously with many soil compounds such as amino acids to form humus

What stimulates soil organic matter decomposition?

- Usual suspects (changes in climate, nutrient availability, aeration)
- Mechanical mixing/plowing
- Plant species change
- C additions “priming the pump”...
- Soil weathering...?

SOIL FORMATION AND AGGREGATION

Where are the microorganisms located in the soil?

Usually the top 2-3 cms.

Commonly found close to root surfaces, in dead roots, on soil particles or amongst aggregates of soil particles.

Soils that are clayey have many bacteria because these soils have lot of small pores.

Sandy soil is less suitable habitat.

Soil aggregates and soil pores

What are soil aggregates and soil pores?

Soil aggregates: Clumps of soil particles that are held together by moist clay, organic matter, gums (from bacteria and fungi) and by fungal hyphae.

Vary in size from 2 microns to 2 mm.

Soil pores: spaces between soil particles and soil aggregates.

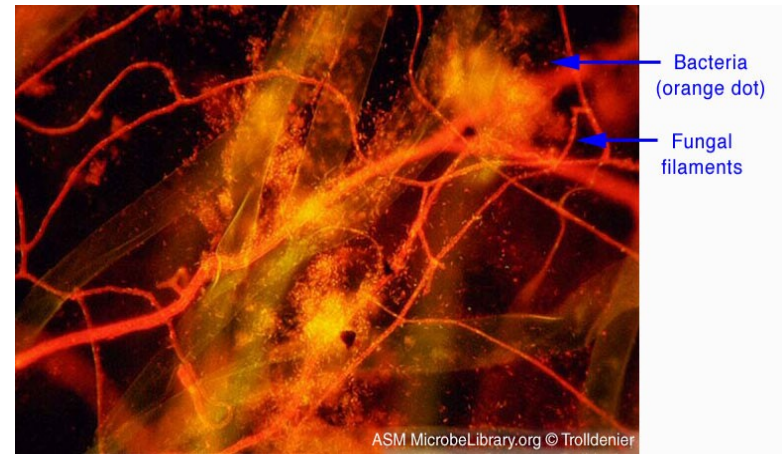
They can be full of air or have water in them.

Very essential for storing air, water, microbes, nutrients and organic matter.

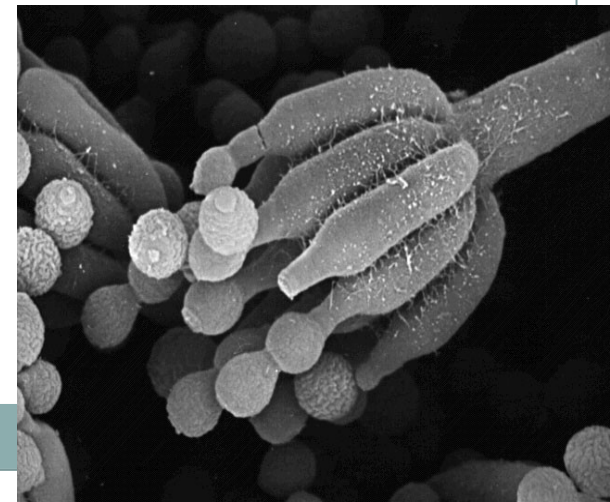
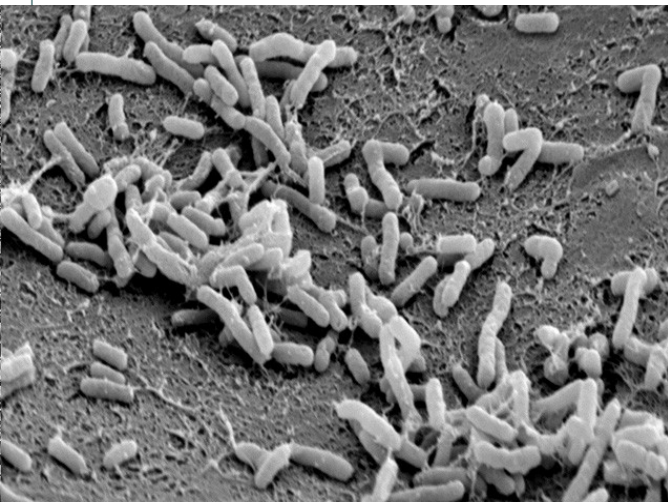
Bacterial/ Fungal role in the formation of soil aggregate

Two ways

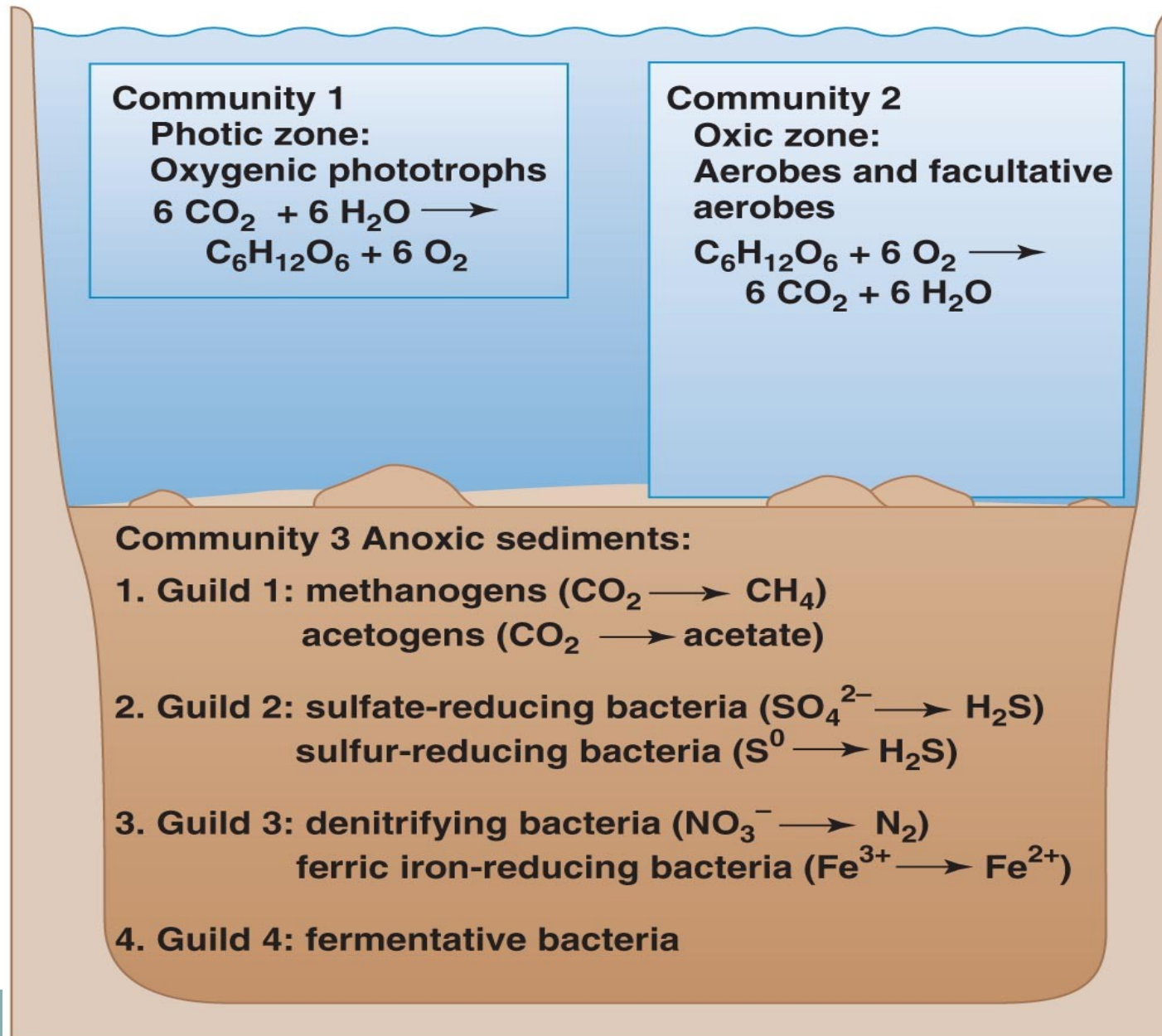
- Using bacterial polysaccharide as a gum
- Through electrostatic charge between bacteria and clay surfaces
- Fungi help form soil aggregates when fungal hyphae enmesh and cross link soil particles.
- Mycorrhizae are very important in this process.



Biogeochemical Cycles and soil microbes



Water body



Nutrient Requirements for Plant Growth

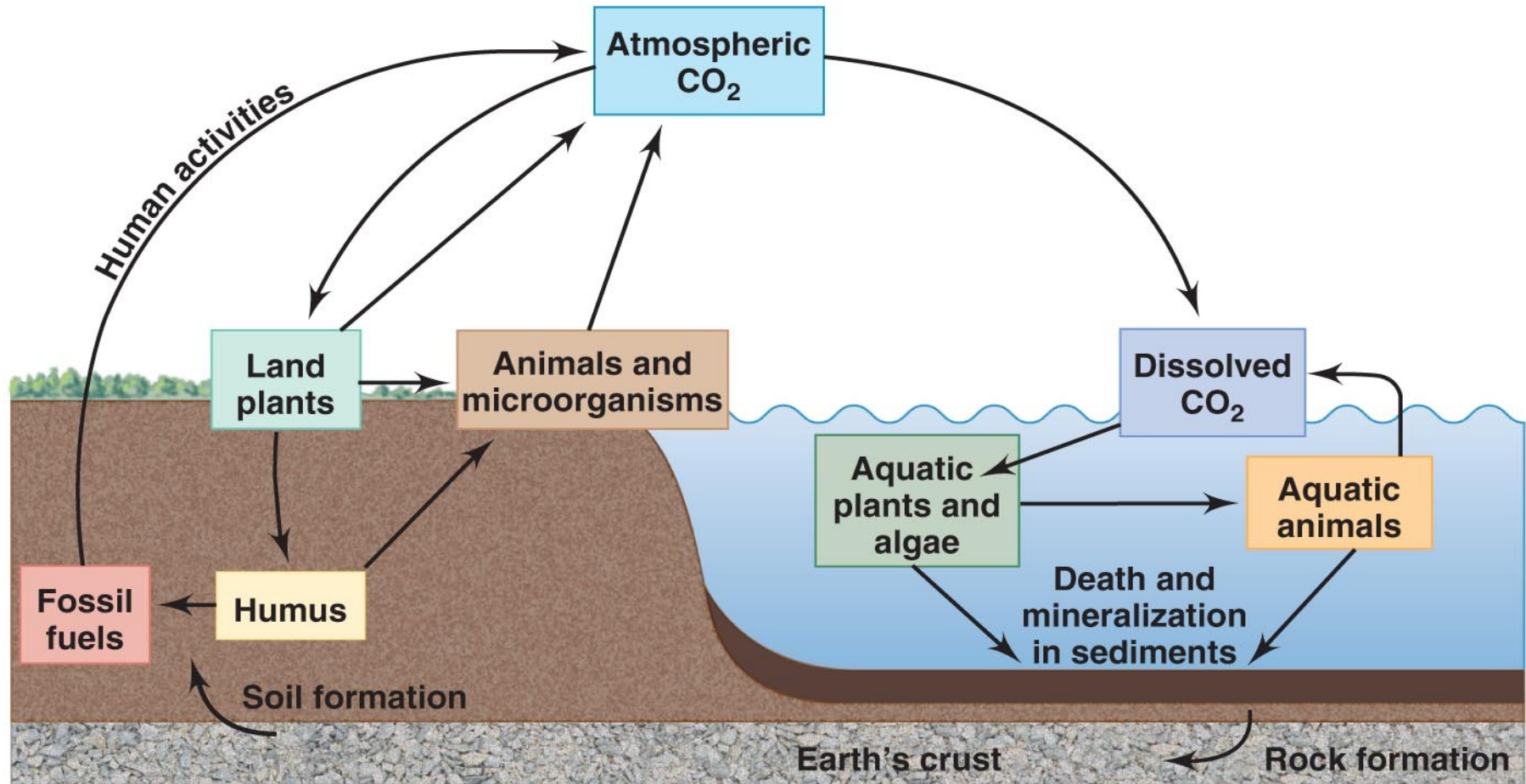
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- **Taken up in gaseous form, Oxygen (O_2), Carbon CO_2 , and from roots - Water (H_2O).**
 - Derived from water and carbon dioxide
- **Rest are taken up from soil solutions**
 - Macro-nutrients -Nitrogen (N), Phosphorous (P), Potassium (K),
 - Calcium (Ca), Magnesium (Mg), Sulfur (S)
 - Micro-nutrients - Boron (B), Copper (Cu), Iron (Fe), Manganese (Mn), Molybdenum (Mo), Zinc (Zn)

Biogeochemical Cycling

- **The cycling of nutrients through ecosystems via food chains and food webs, including the exchange of nutrients between the biosphere and the hydrosphere, atmosphere and geosphere (e.g., soils and sediments)**

The Carbon Cycle



Carbon is cycled through all of Earth's major carbon reservoirs (atmosphere, land, oceans, sediments, rocks, and biomass)

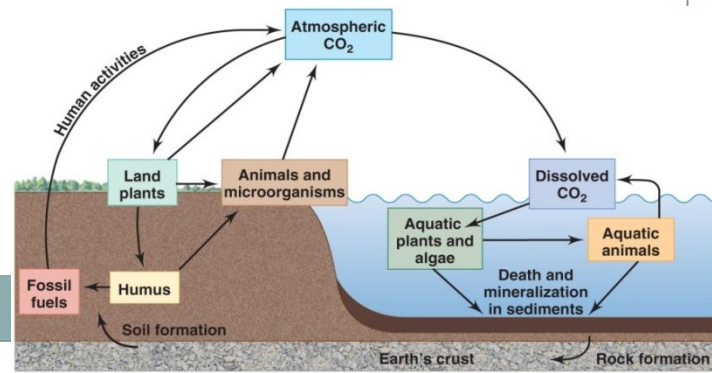
The Carbon Cycle

CO₂ in the atmosphere is the most rapidly transferred carbon reservoir

CO₂ is fixed primarily by photosynthetic land plants and marine microbes (autotrophs fix carbon dioxide)

CO₂ is returned to the atmosphere by respiration of animals and chemoorganotrophic microbes as well as by the activities of humans.

Microbial decomposition is the largest source of CO₂ released to the atmosphere



Carbon Cycle

Autotrophs fix carbon dioxide into organic molecules

Chemoheterotrophs eat autotrophs and each other (This moves the carbon atoms around the food chain)

As carbon containing organic compounds are used for energy (respiration) carbon dioxide is released into the atmosphere to start the cycle over again.

Also: when organisms die their remains are acted upon by bacteria and fungi (decomposers). Decomposition oxidizes the organic remains and carbon dioxide is liberated to re-enter the cycle.

Methanogenesis and Syntrophy

Most methanogens reduce CO_2 to CH_4 with H_2 as an electron donor; some can reduce other substrates to CH_4 (e.g., acetate)

Methanogens may team up with partners (syntrophs) that supply them with necessary substrates

Syntrophy: A process whereby 2 or more microbes cooperate to degrade a substance neither can degrade alone

Methylotrophs are a diverse group of microorganisms that can use reduced one-carbon compounds, such as methanol or methane as the carbon source for their growth.

Some methylotrophs can degrade the methane and in this case they are called methanotrophs.

The methanotroph *Methylococcus capsulatus* used to degrade methane and other pollutants.

Examples of methylotrophs:

Genus [Methanosarcina](#)

[Methylococcus capsulatus](#)

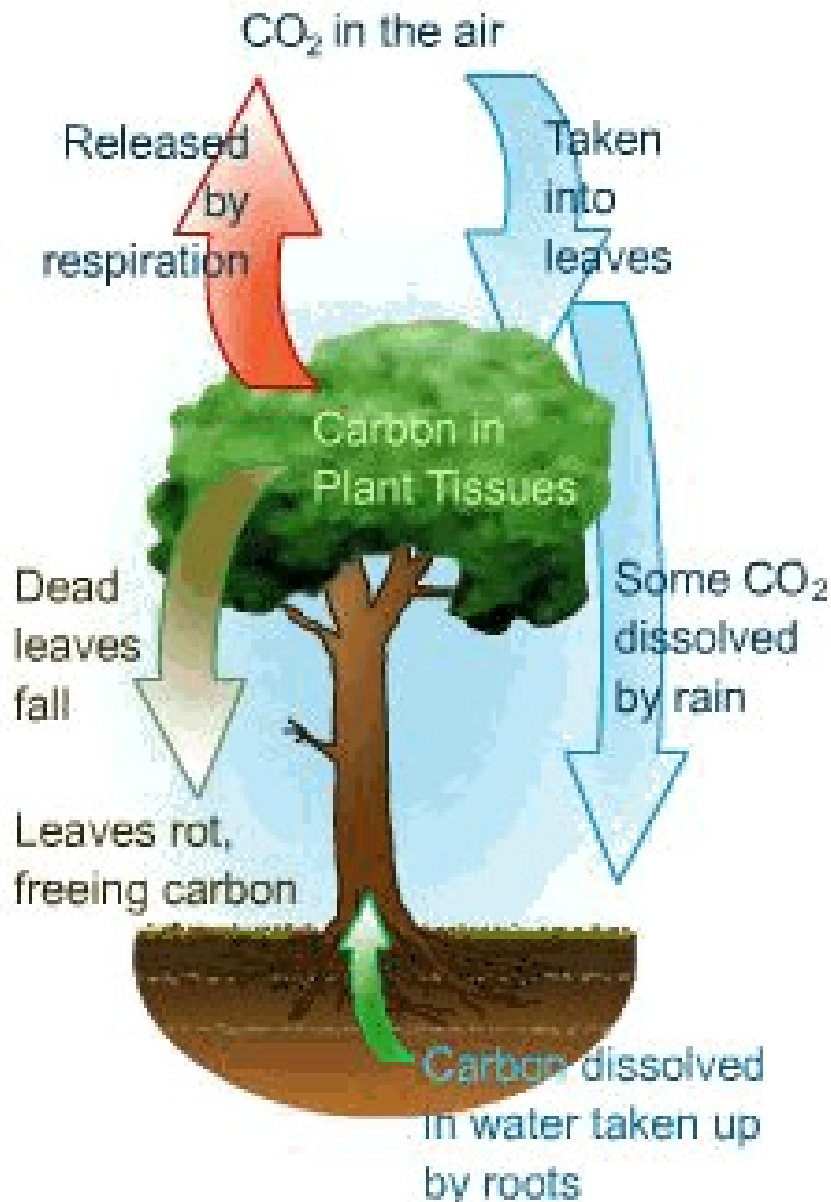
Carbon cycling

Methane (CH_4)

is an end product of anaerobic ecosystems

is a greenhouse gas; traps 21X as much heat per molecule as CO_2

is consumed by some methylootrophs



Nitrogen Cycle

Nitrogen is essential for amino acids, nucleic acids, etc.

80% of atmosphere is molecular nitrogen N_2

For plants to use nitrogen it must be fixed (combined with other elements). Only a few species of bacteria can fix nitrogen!!

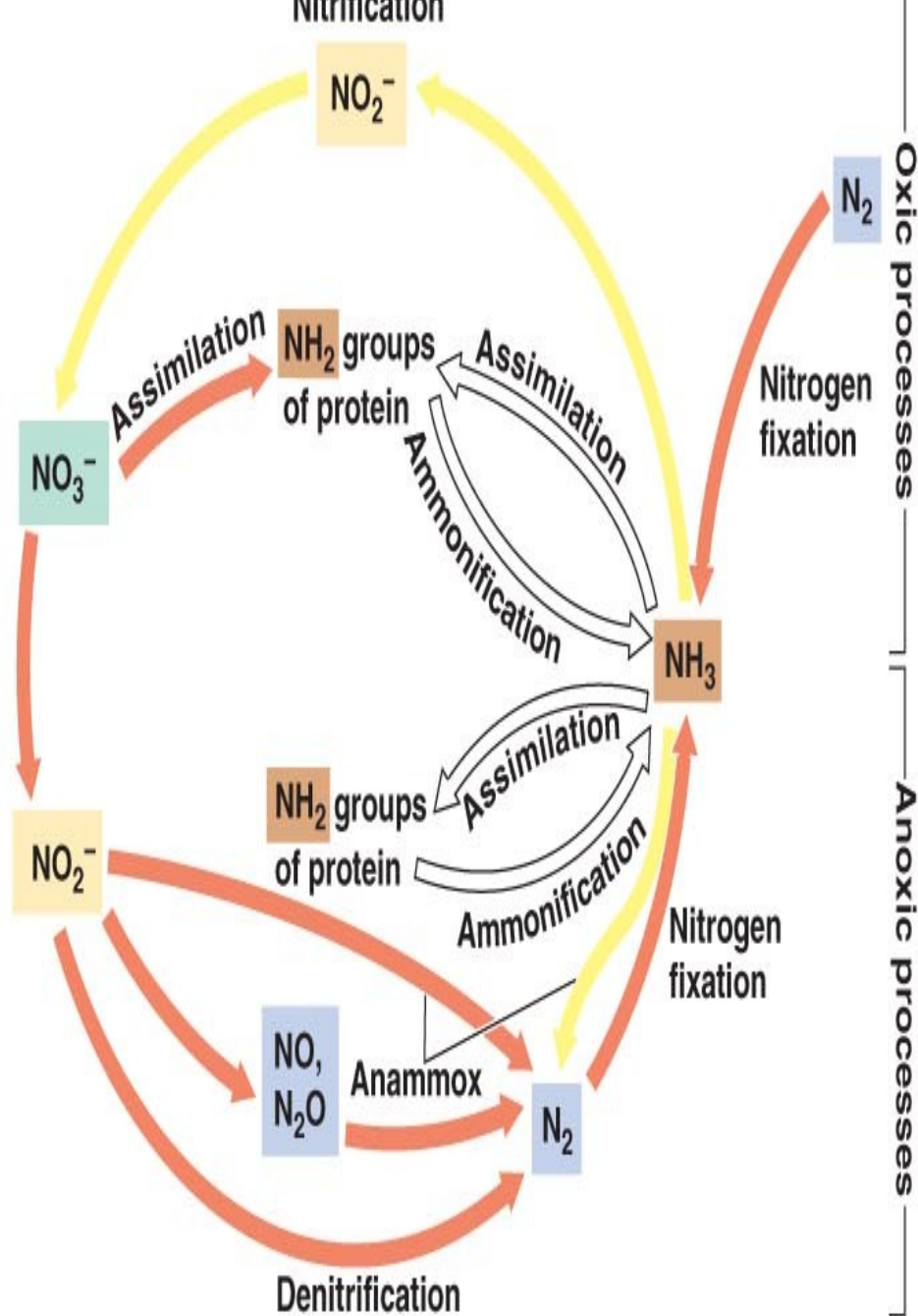
Nitrogen fixation: nitrogen gas is converted to ammonia (requires enzyme nitrogenase)

Free-living nitrogen-fixing bacteria

Symbiotic nitrogen-fixing bacteria

Key Processes and Prokaryotes in the Nitrogen Cycle

Processes	Example organisms
Nitrification ($\text{NH}_4^+ \rightarrow \text{NO}_3^-$)	
$\text{NH}_4^+ \rightarrow \text{NO}_2^-$	<i>Nitrosomonas</i>
$\text{NO}_2^- \rightarrow \text{NO}_3^-$	<i>Nitrobacter</i>
Denitrification ($\text{NO}_3^- \rightarrow \text{N}_2$)	<i>Bacillus, Paracoccus, Pseudomonas</i>
N_2 Fixation ($\text{N}_2 + 8\text{H} \rightarrow \text{NH}_3 + \text{H}_2$)	
Free-living	
Aerobic	<i>Azotobacter</i> Cyanobacteria
Anaerobic	<i>Clostridium</i> , purple and green bacteria
Symbiotic	<i>Rhizobium</i> <i>Bradyrhizobium</i> <i>Frankia</i>
Ammonification (organic-N $\rightarrow \text{NH}_4^+$)	Many organisms can do this
Anammox ($\text{NO}_2^- + \text{NH}_3 \rightarrow 2\text{N}_2$)	<i>Brocadia</i>



The Nitrogen Cycle

Ammonification

Microbes decompose proteins to amino acids. The amino acids can undergo deamination which produces ammonia (NH_3)

Ammonia produced by nitrogen fixation or ammonification can be assimilated into organic matter or oxidized to nitrate

Nitrification the oxidation of ammonia to produce nitrate.

Plants use nitrate as their principle source of nitrogen

Nitrobacter and *Nitrosomonas* are genera of nitrifying bacteria

Nitrogen Cycle

- Denitrification is the reduction of nitrate (NO_3) to gaseous nitrogen products (N_2O and N_2) and is the primary mechanism by which N_2 is produced biologically
 - *Pseudomonas* spp. are important bacteria in the denitrification of soils
- Anammox is the anaerobic oxidation of ammonia to N_2 gas
- Denitrification and anammox result in losses of nitrogen from the biosphere

Biological Nitrogen Fixation

Occurs only in prokaryotes:

Bacteria (e.g. *Rhizobium*, *Frankia*)

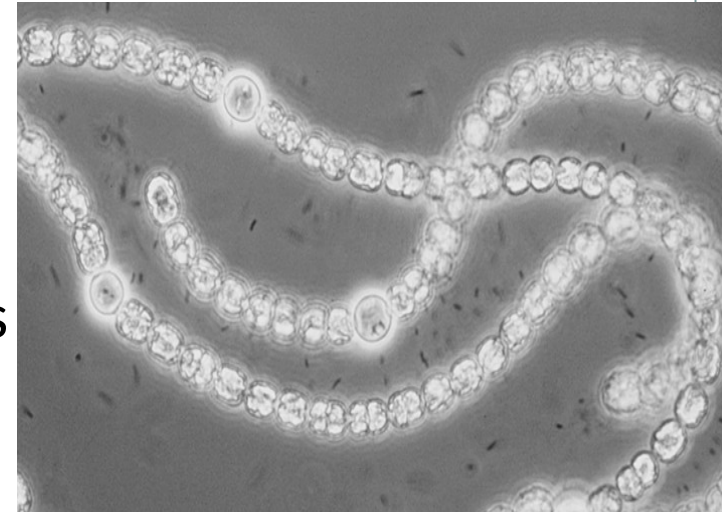
Cyanobacteria (e.g. *Nostoc*, *Anabaena*)

Free-living in soil/water –
heterocysts

Symbiotic with plants – root nodules

Loose association with plants

Mutualism



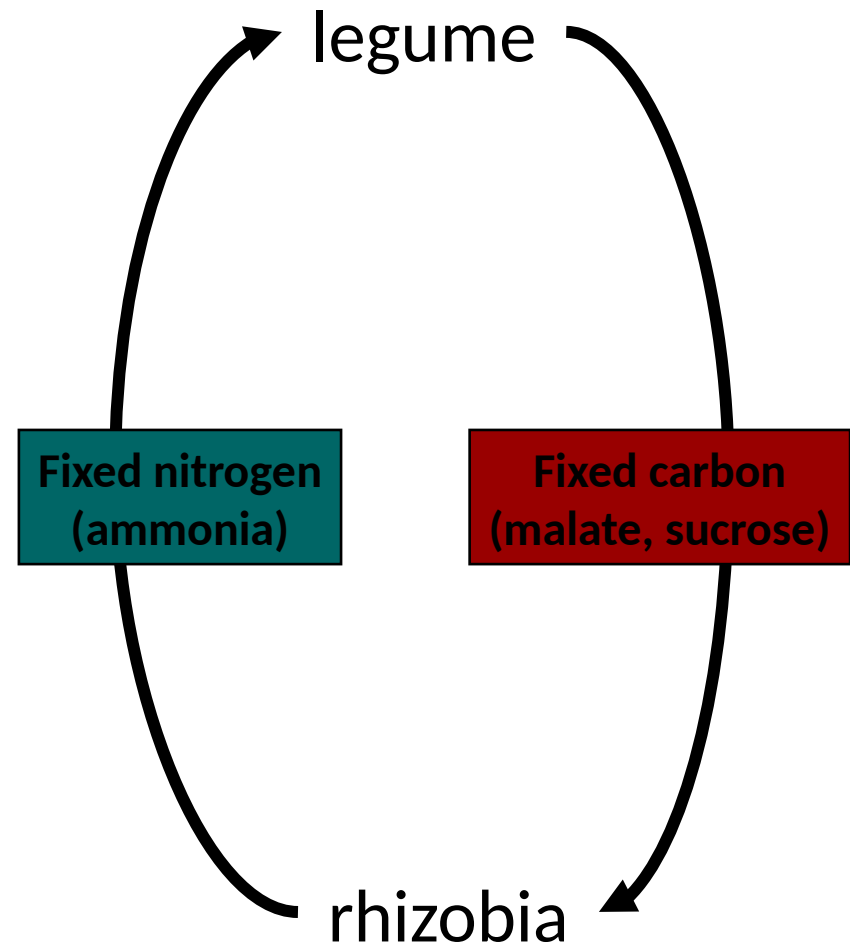
Anabaena with heterocysts

Prokaryote receives carbohydrates

Plant may allocate up to 30% of its C to the symbiont

Plant provides anaerobic site – nodules

Plant receives N



1) Legumes (Fabaceae)

Widespread

bacteria = e.g., *Rhizobium* spp.

N₂-fixing symbionts form root “nodules”

– anaerobic sites that “house” bacteria

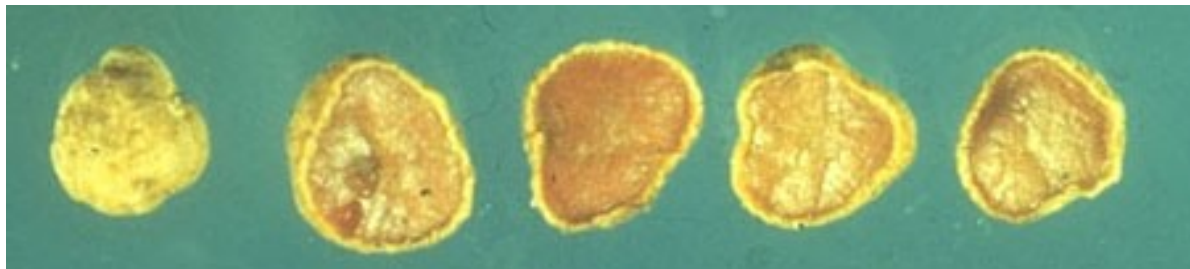


alpine clover



Symbionts regulate O_2 in the nodule with leghemoglobin

Different part synthesized by the bacteria and
legume



Cross-section of nodules of soybean nodules

Rhizobium-legume symbioses

Host plant

Bacterial symbiont

Alfalfa

Rhizobium meliloti

Clover

Rhizobium trifolii

Soybean

Bradyrhizobium japonicum

Beans

Rhizobium phaseoli

Pea

Rhizobium leguminosarum

Sesbania

Azorhizobium caulinodans

Both plant and bacterial factors determine specificity

Part of a clover root system bearing naturally occurring nodules of *Rhizobium*. Each nodule is about 2-3 mm long



Clover root nodules showing two partly crushed nodules (arrowheads) with pink-colored contents. This color is caused by the presence of the pigment leghemoglobin - a unique metabolite of this type of symbiosis. Leghemoglobin is found only in the nodules and is not produced by either the bacterium or the plant when grown alone.



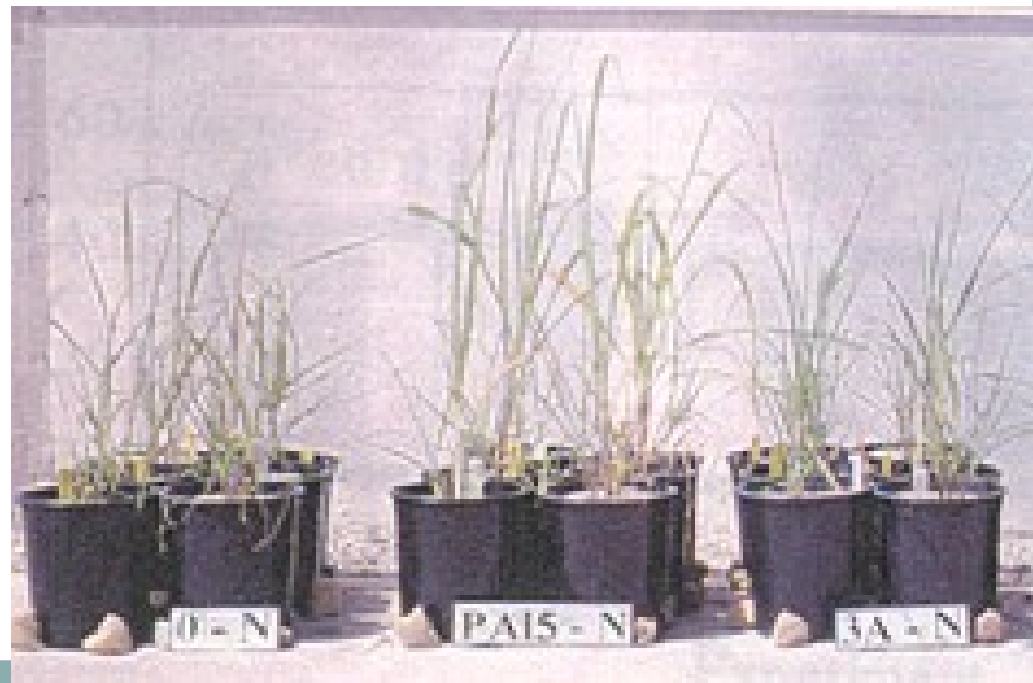
Plant-associated nitrogen fixation: the endophytes and epiphytes



Acetobacter diazotrophicus

Lives as an endophyte of sugarcane and various other monocots and some dicots

On sugarcane



Another cyanobacterium on the palm *Welfia regia* in an epiphyllic relationship



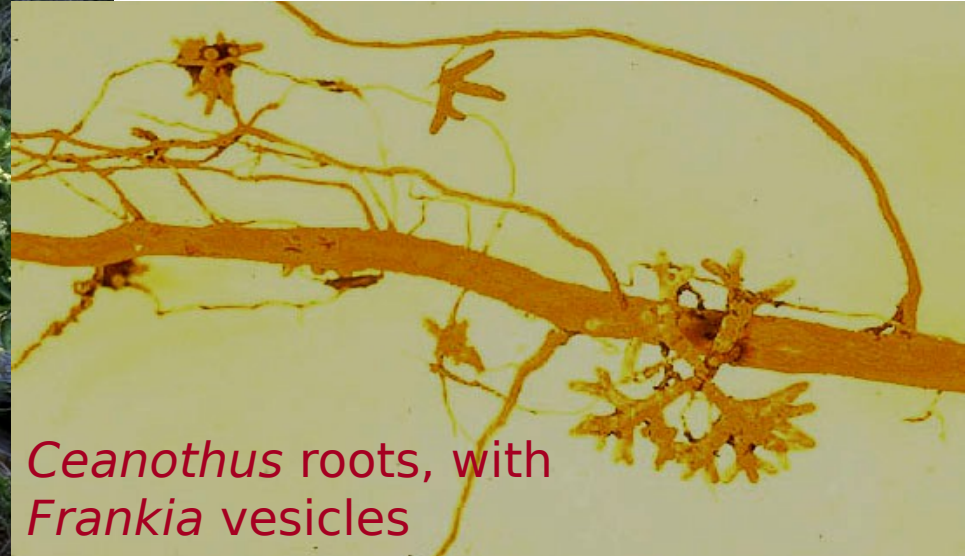
It is believed that these bacteria transfer some % of fixed N to the plants through the leaf surfaces

2) Non-legume symbiotic plants –

Actinorhizal”= associated with actinomycetes (N_2 -fixing bacteria)

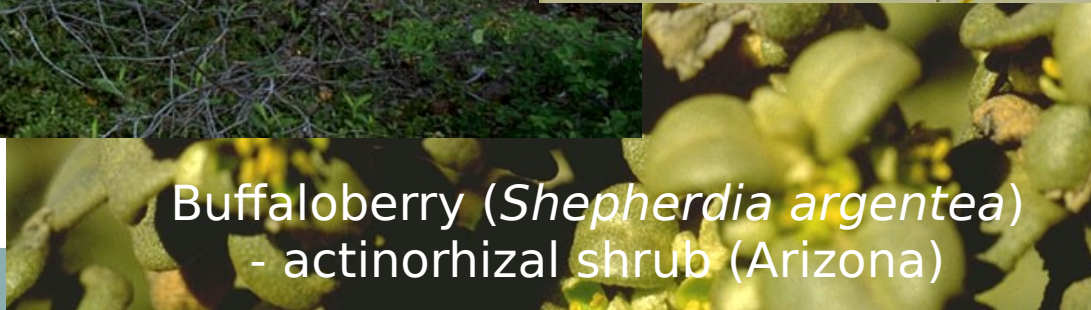
- genus *Frankia*
- Usually woody species – e.g. Alders, *Ceanothus*

Ceanothus velutinus - snowbrush

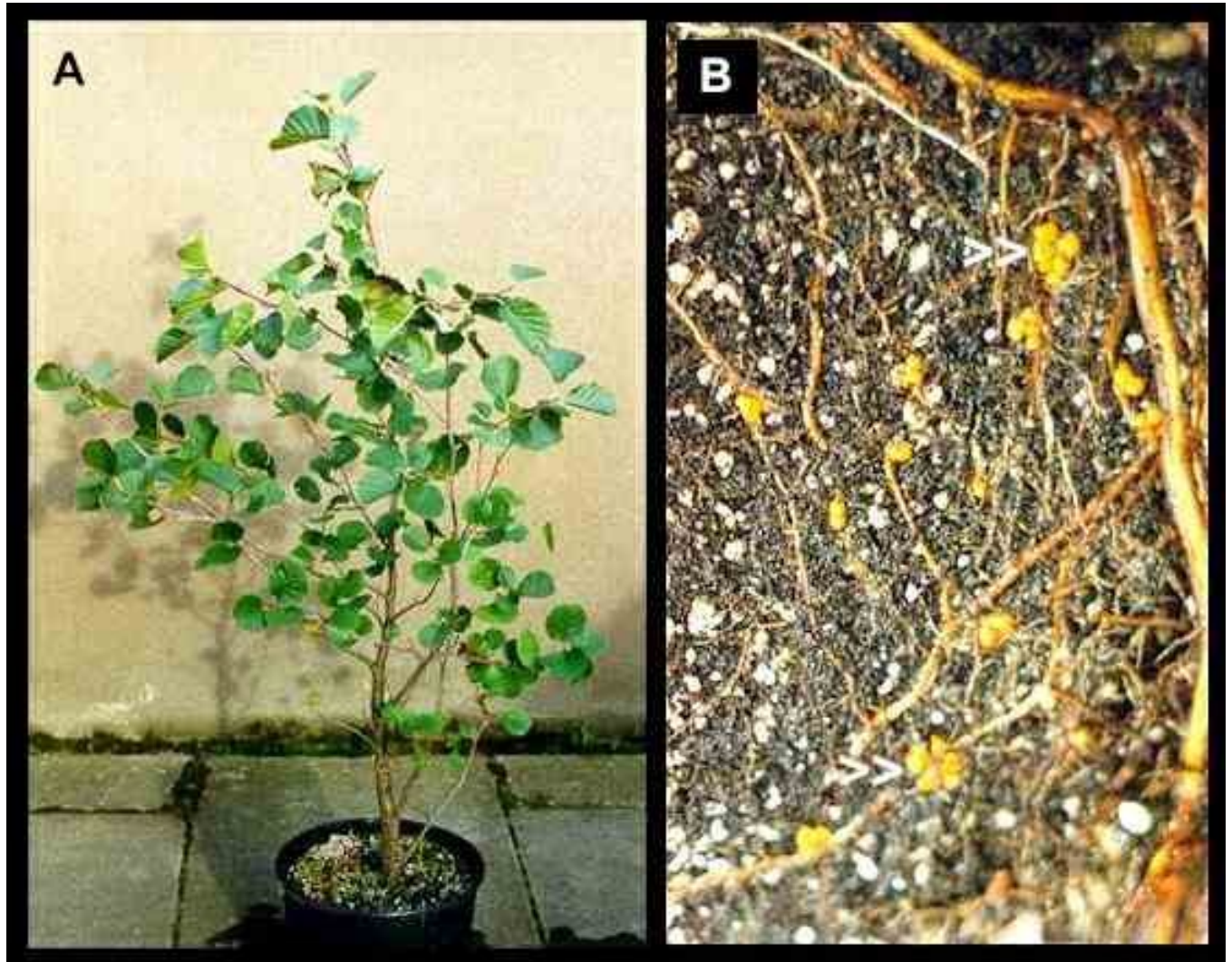


Ceanothus roots, with
Frankia vesicles

Buffaloberry (*Shepherdia argentea*)
- actinorhizal shrub (Arizona)



Alder and the other woody hosts of *Frankia* are typical pioneer species that invade nutrient-poor soils. These plants probably benefit from the nitrogen-fixing association, while supplying the bacterial symbiont with photosynthetic products.





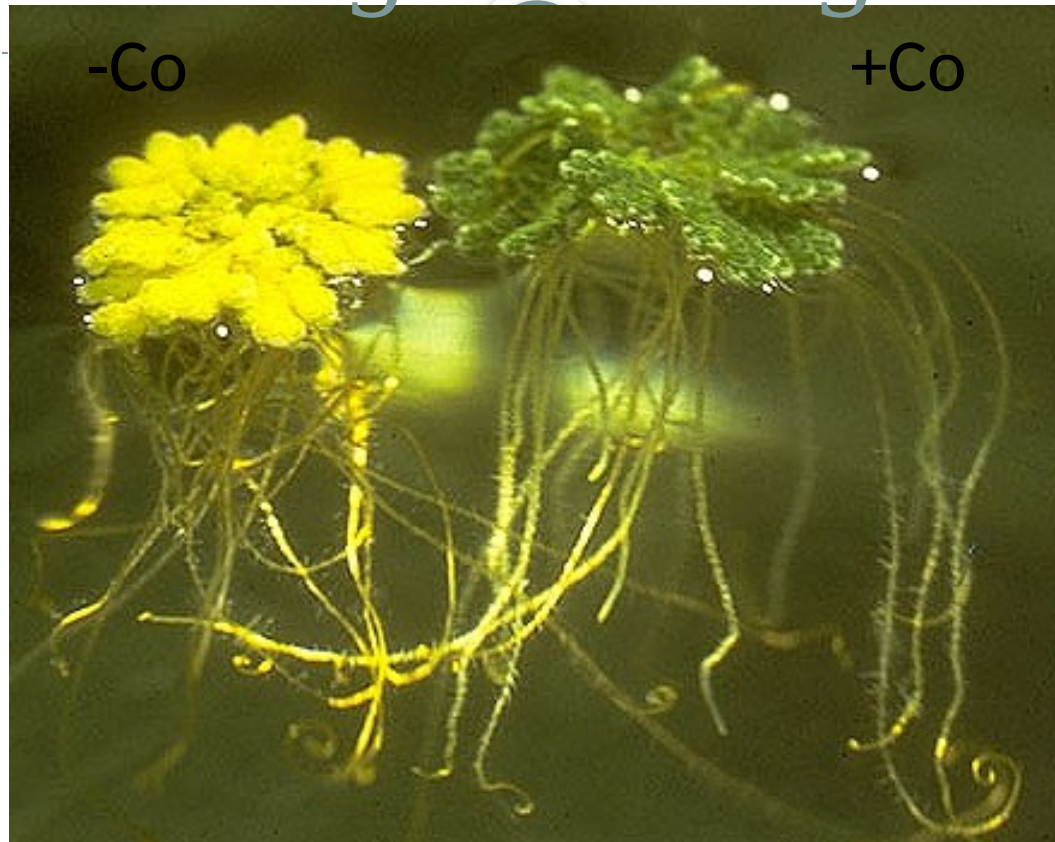
Cyanobacteria ; Symbiotic cyanobacteria,
with fungi- lichens

With aquatic fern *Azolla*- *Anabaena azollae*

With a Cycad-*Anabaena cycadarum*

Azolla harbouring *Anabaena* is an important
green manure for rice cultivation

A nitrogen-fixing fern



The aquatic fern *Azolla* is the only fern that can fix nitrogen. It does so by virtue of a symbiotic association with a cyanobacterium (*Anabaena azollae*).

The Sulfur Cycle

- H_2S (Hydrogen sulfide...reduced form of sulfur) is an energy source for some autotrophs
 - Photoautotrophs (green and purple bacteria groups)
 - Chemoautotrophs (*Beggiatoa* and *Thiobacillus*)
- Oxidation of sulfur
 - H_2S -----> elemental sulfur (S) -----> SO_4
- Sulfate (SO_4) is assimilated by plants and bacteria into proteins
- Decomposition by microbes releases sulfur (dissimilation) as H_2S to reenter the cycle

Redox Cycle for Sulfur

Key Processes and Prokaryotes in the Sulfur Cycle

Processes	Example organisms
Sulfide/sulfur oxidation ($\text{H}_2\text{S} \rightarrow \text{S}^0 \rightarrow \text{SO}_4^{2-}$)	
Aerobic	Sulfur chemolithotrophs (<i>Thiobacillus</i> , <i>Beggiatoa</i> , many others)
Anaerobic	Purple and green phototrophic bacteria, some chemolithotrophs
Sulfate reduction (anaerobic) ($\text{SO}_4^{2-} \rightarrow \text{H}_2\text{S}$)	<i>Desulfovibrio</i> , <i>Desulfobacter</i>
Sulfur reduction (anaerobic) ($\text{S}^0 \rightarrow \text{H}_2\text{S}$)	<i>Desulfuromonas</i> , many hyperthermophilic <i>Archaea</i>
Sulfur disproportionation ($\text{S}_2\text{O}_3^{2-} \rightarrow \text{H}_2\text{S} + \text{SO}_4^{2-}$)	<i>Desulfovibrio</i> , and others
Organic sulfur compound oxidation or reduction ($\text{CH}_3\text{SH} \rightarrow \text{CO}_2 + \text{H}_2\text{S}$) ($\text{DMSO} \rightarrow \text{DMS}$)	Many organisms can do this
Desulfurylation (organic-S $\rightarrow \text{H}_2\text{S}$)	Many organisms can do this

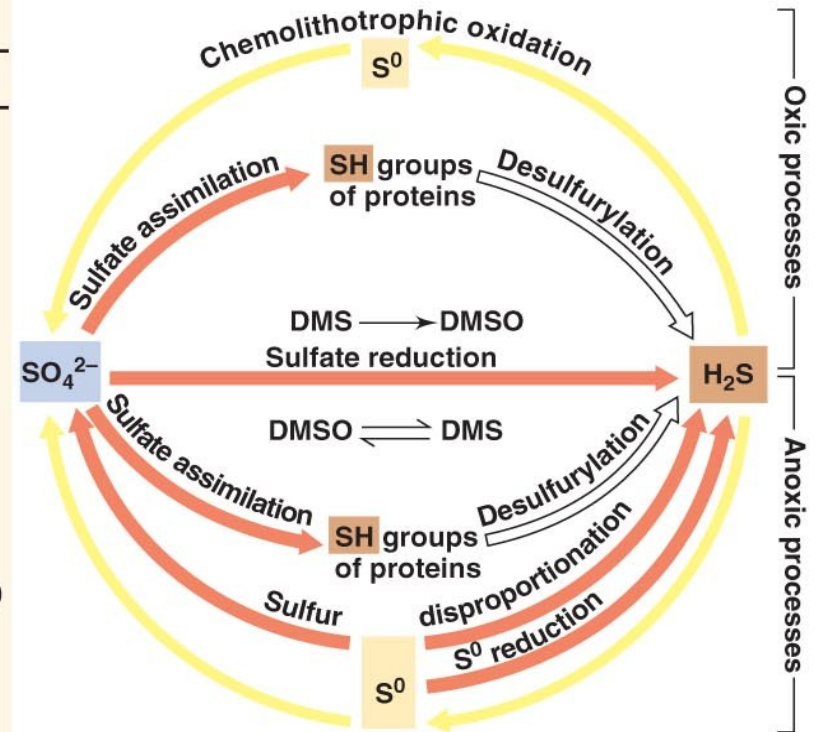


Figure 24.6



Thiobacillus are colorless, rod-shaped, Gram-negative bacteria with polar flagella. They possess an iron oxidase, which allows them to metabolize metal ions such as ferrous iron:



Thiobacillus are strictly aerobic bacteria.



Thiobacillus ferrooxidans Or *Acidithiobacillus ferrooxidans* is the most common type of bacteria in mine waste piles. This organism is acidophilic (acid loving), and increases the rate of pyrite oxidation in mine tailings piles and coal deposits. It oxidizes iron and inorganic sulfur compounds. The oxidation process can be harmful, as it produces sulfuric acid, which is a major pollutant.



Thiobacillus ferrooxidans.

Thiobacillus

Strictly aerobic

Gram-negative

Rod shaped

Polarly flagellated

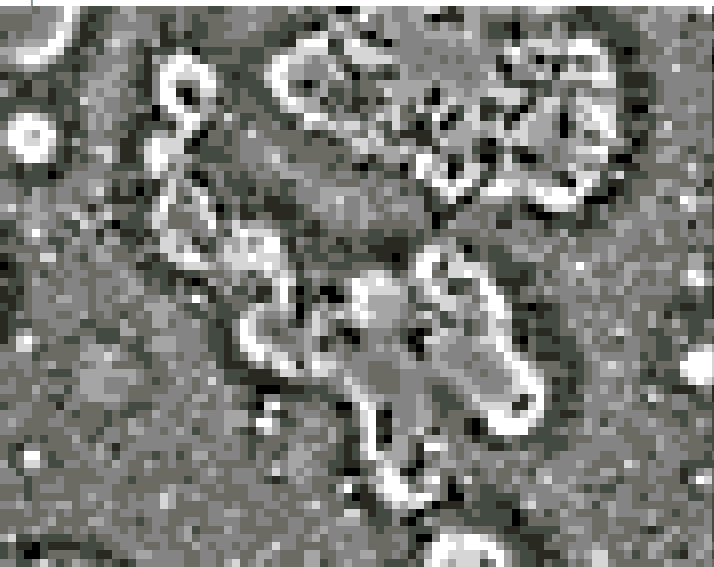
Non-spore forming

Best growth at 25-35° C

Obligate autotrophs

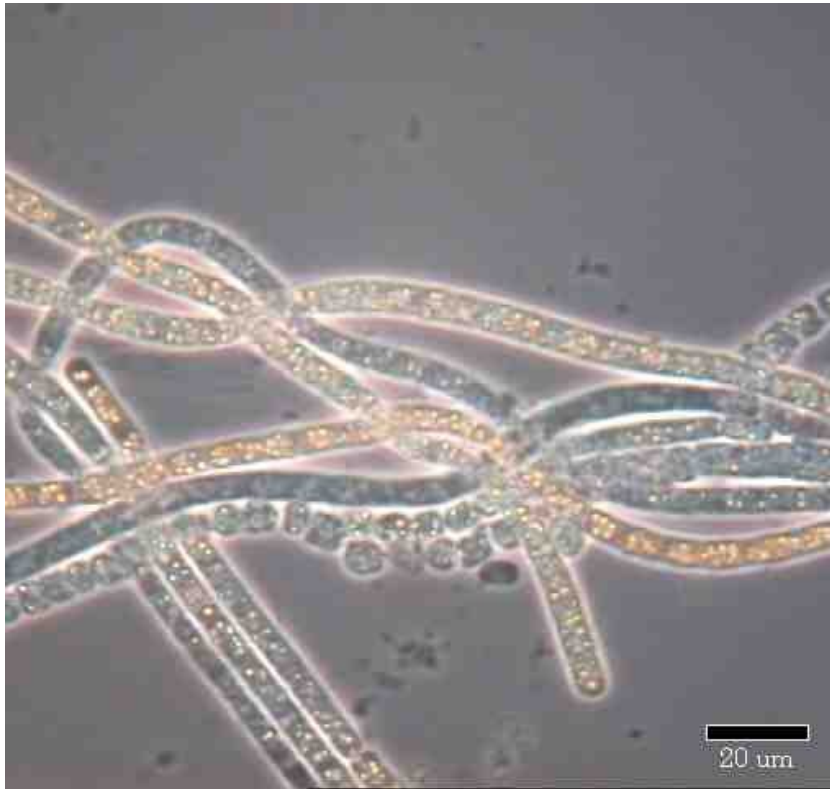
Grow best at neutral pH

(Some spp. are able to live in highly acidic environments)



cells of thiobacilli growing in a "floc" of cellular material and sulfur in water.

Beggiatoa



Aerobic or microaerophilic

Gram-negative

Disk shaped, filament forming

Motile by gliding

Non-spore forming

Growth occurs between 0-40 °C

Chemoorganotrophic and facultatively autotrophic

Sulfolobus



Strictly aerobic

Archaeabacteria

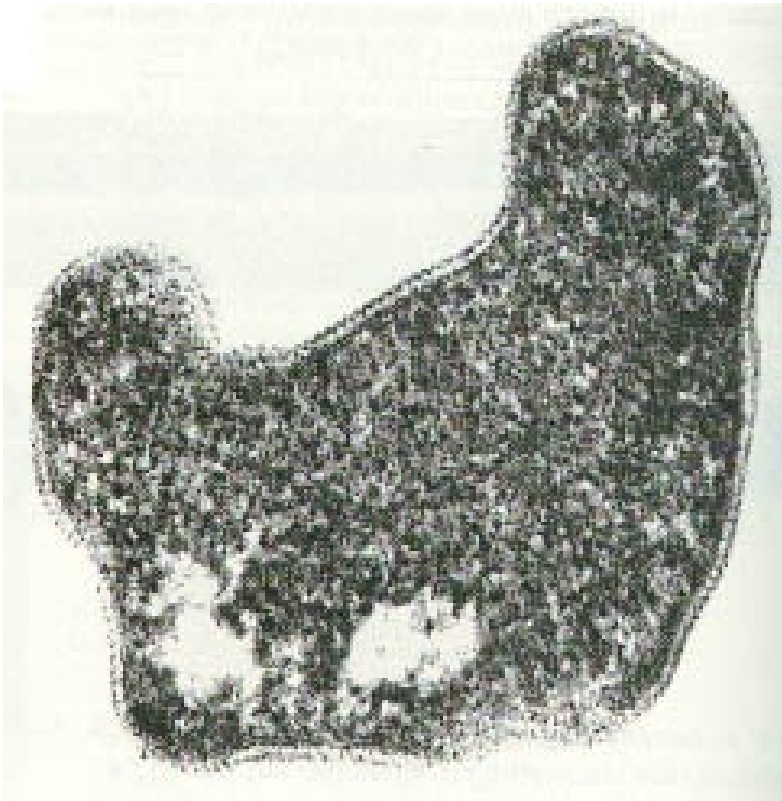
Irregular shape, filament forming

Having pilus or pseudopodia

Found in pH 1 - 6, cannot survive above pH 7

Temperature range is form 55 - 87° C

Capable of lithotrophic and organotrophic growth





Sulfolobus can grow either lithoautotrophically by oxidizing sulfur, or chemoheterotrophically using sulfur to oxidize simple reduced carbon compounds. Heterotrophic growth has only been observed, however, in the presence of oxygen. The principle metabolic pathways are a glycolytic pathway, a pentose phosphate pathway, and the TCA cycle.



S. solfataricus has been found in different areas including Yellowstone National Park, Mount St. Helens, Iceland, Italy, and Russia to name a few.

Sulfolobus is located almost wherever there is volcanic activity. They thrive in environments where the temperature is about 80°C with a pH at about 3 and sulfur present.

Another species, *S. tokodaii*, has been located in an acidic spa in Beppu Hot Springs, Kyushu, Japan. Sediments from ~90m below the seafloor on the Peruvian continental margin are dominated by community is sedimentary archaea.

Sulfolobus as a viral host



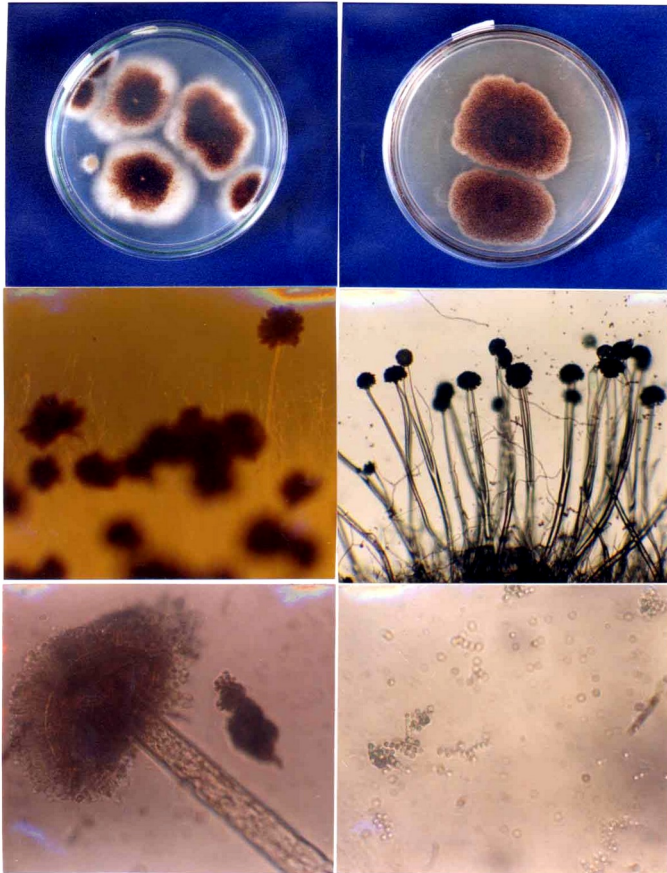
Lysogenic viruses infect *Sulfolobus* for protection. The viruses cannot survive in the extremely acidic and hot conditions that *Sulfolobus* lives in, and so the viruses use *Sulfolobus* as protection against the harsh elements. This relationship allows the virus to replicate inside the archaea without being destroyed by the environment.

Sulfolobus proteins are of interest for biotechnology and industrial use due to their thermostable nature. One application is the creation of artificial derivatives from *S. acidocaldarius* proteins, named affitins.

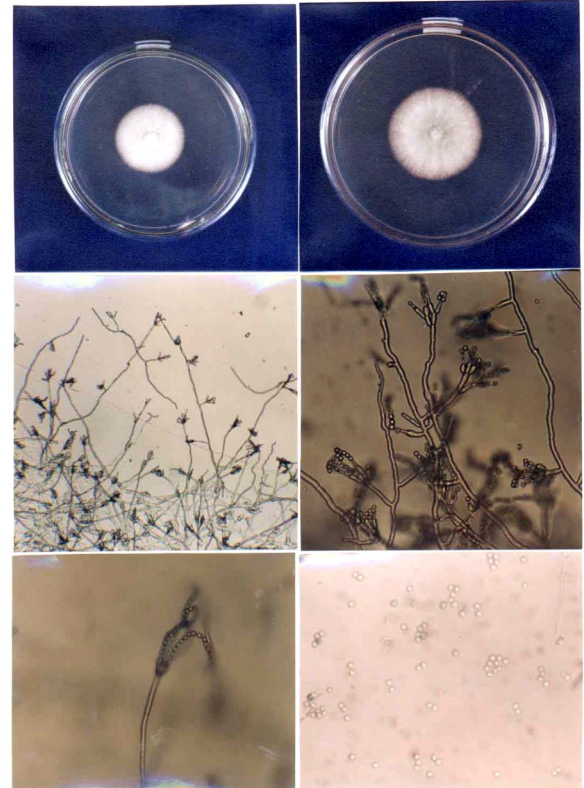
Intracellular proteins are not necessarily stable at low pH however, as *Sulfolobus* species maintain a significant pH gradient across the outer membrane.

S. tokodaii is known to oxidize hydrogen sulfide to sulfate intracellularly.

SULFUR OXIDIZING FUNGI



Aspergillus japonicus



Penicillium nalgiovense

The purple sulfur bacteria



Can use Hydrogen sulfide as a photosynthetic electron donor.

Oxidation of hydrogen sulfide produces sulphur. In the genus of *Ectothiorhodospira* **sulphur deposition is extracellular.**

All the other purple sulfur bacteria sulfur is deposited within the cells as oily inclusions.

Examples:

Thiospirillum, *Thiocystis*, *Thiocapsa*,
Chromatium

GREEN SULFUR BACTERIA



- Photolithotrophic and Anaerobic
- Straight or curved rods to star-like extrusions, some spiral forms
- Gram negative
- Non motile
- Colonies are green, and brown (if carotenoids are present)
- Nitrogen fixation
- Non spore forming

Chlorobium tepidum is a thermophilic green sulfur bacteria originally isolated from a New Zealand hot spring. This type of bacterium has special light-harvesting complexes called chlorosomes that contain bacteriochlorophylls and carotenoids.

Chlorobium grows in dense mats over hot springs as well as in other warm muds and bodies of water that contain sufficient hydrogen sulfide.



The green bacteria are strictly anaerobic and obligatory photosynthetic. They can use hydrogen sulfide as an electron donor to the reduction of carbon dioxide. The sulfur formed as an intermediate in oxidation of sulfide to sulphate is always deposited outside the cells.

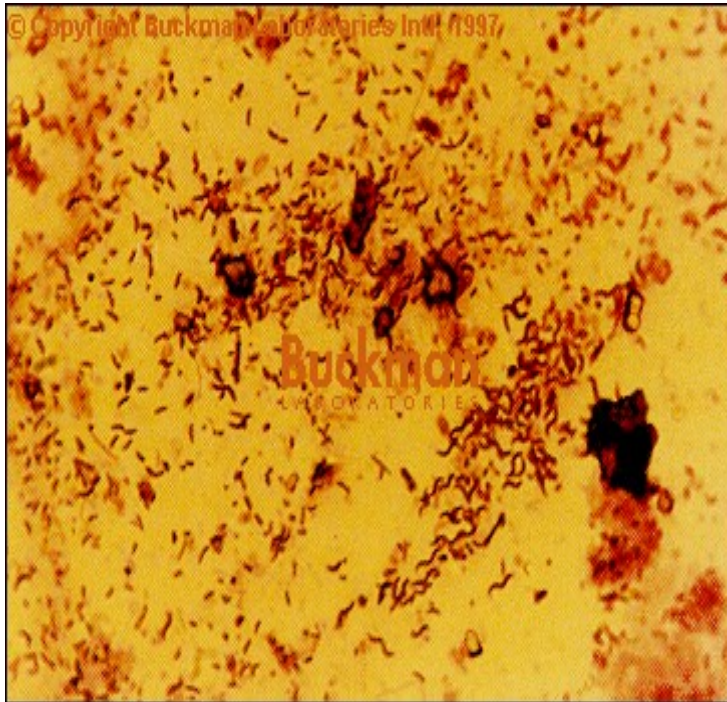
REDUCTION OF OXIDIZED S



Environmental conditions:

- Anaerobic conditions
(All O_2 and NO_3^- gone)
- Available organic C exist
- Desulfuring microorganisms exist
(*Desulfovibrio*)

Desulfovibrio desulfuricans



- Obligate anaerobes
- Curved rod-shaped cells
- Non-spore forming
- Gram negative
- Polarly flagellated

Sulfate reducers



The rod-shaped bacterium is identified as anaerobically growing, motile organism. *Desulfovibrio* is known as a sulfate reducing bacterium, which has put it to the forefront of biological research. Because of its metal corroding ability, which has consequently led to numerous health and safety concerns in industry.

The organism however also shows potential for bioremediation, in that it may undergo anaerobic conversion of pollutants in the soil.

Phosphorus Cycle

- Phosphorus exists mainly as phosphate (PO_4) ions
- The phosphorus cycle mainly involves changes from soluble to insoluble forms of phosphate.
- Microbes (such as *Thiobacillus*) may produce acids that solubilize the phosphate and make it available for plants and other microbes.

Plant-Mycorrhizae Mutualism

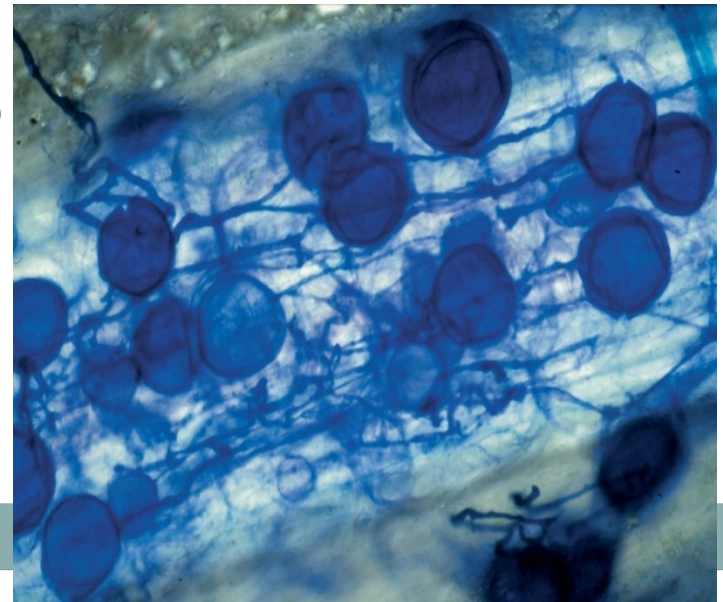
- mycorrhizal fungi provide phosphorus to plants (**nutritional**)
- other part of mutualism is **energetic** - plants provide carbohydrates

Widespread –

Occurs ~80% angiosperm spp

All gymnosperms

Sometimes an obligate relationship



Ectomycorrhizae

fungus covers the ends of young roots & only penetrates the cell wall of the cortex

common in evergreen trees/shrubs



deciduous tree species: *Fagus*, *Betula*, *Quercus*, *Tilia*, *Populus*, *Salix* and *Castanea*

absorb phosphate, ammonium, and zinc from the soil

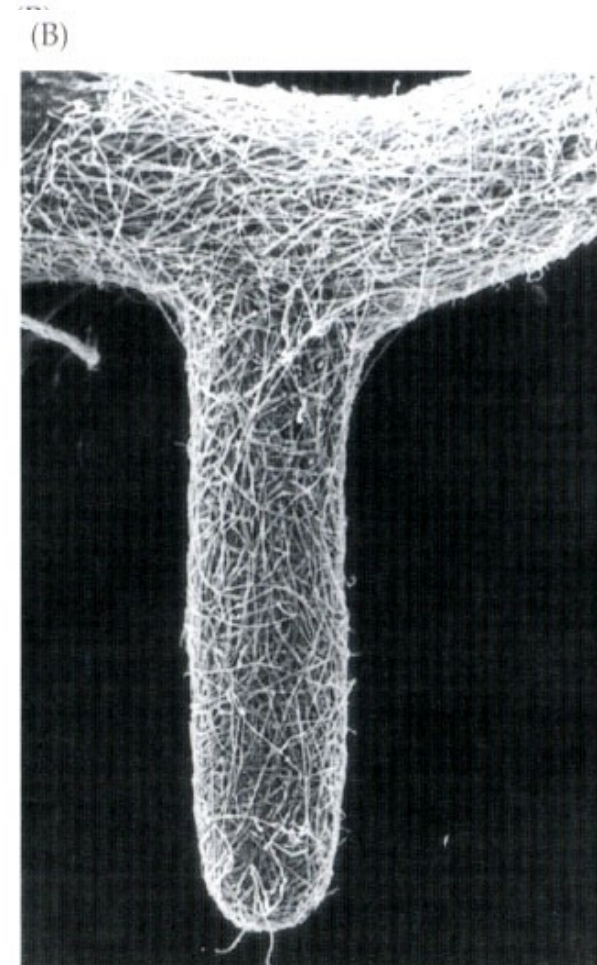
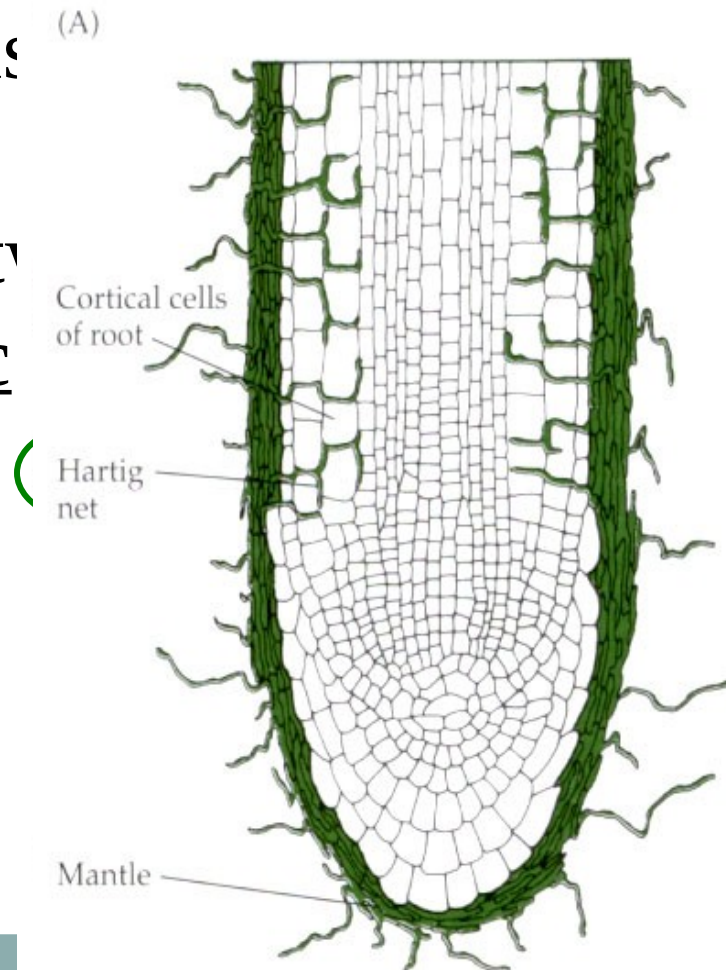
tend to be host-specific

B. Major groups of mycorrhizae:

1) Ectomycorrhizae –

Fungus forms
(mantle)

Grows in betw
– apoplastic



Endomycorrhizae



invade plant roots and develop within the plant
vesiculararbuscular mycorrhizae (VAM)

commonly on deciduous trees, annual crops, &
other herbaceous plants

hyphae grow inter & intracellularly within the
root

uptake of phosphate and zinc

typically not host-specific

Endomycorrhizae -

Fungus penetrates cells of root

example is arbuscular mycorrhizae

(AM)

Found in both herbaceous & woody plants

Arbuscule in plant cell

